Consensus-Supervised U-Net Fold Ensemble for Solar Filament Instance Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. We train U-Nets with an ImageNet-pretrained EfficientNet-B3 encoder on native-resolution crops of the 2048×2048 images, supervising them with *soft consensus* masks that encode the fraction of annotators who marked each pixel, together with two auxiliary targets (a normalised distance-to-boundary map and the contact lines between touching filaments). Five models, one per date-grouped cross-validation fold, are averaged at inference together with flip test-time augmentation; the averaged probability is thresholded and converted into pixel-disjoint instances by connected-component labelling with a minimum-area filter. Post-processing parameters are chosen on out-of-fold predictions against the Panoptic Quality (PQ) metric computed against every annotator set independently. In five-fold cross-validation a single model reaches an out-of-fold PQ of 0.404± 0.004 (1154 annotation sets); the five-model ensemble scores 0.35 on the public Kaggle leaderboard. A ResNet34 variant, a distance-map watershed, contact-line splitting and confidence filtering of components were evaluated and rejected because they did not improve PQ. The code, trained weights and a notebook reproducing the predictions are publicly available.[1]

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[2]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

The task is *instance* segmentation evaluated with Panoptic Quality [6]: every predicted filament counts either as a true positive (IoU $>$ 0.5 with one ground-truth filament), or as a false positive, and each pixel may belong to at most one prediction. Three properties of the data shaped our design. First, the 707 training images carry 1154 annotation sets, because 296 images were annotated independently by two or three annotators who disagree noticeably on filament extent and on faint filaments. Second, filaments are small and thin (median area 1228 px, median bounding box ≈75×70 px in a 2048×2048 frame), so down-sampling destroys the barbs the organizers care about. Third, only 0.9% of the 8199 annotated filaments touch another filament within 3 px, so the instance problem is almost entirely a *semantic* segmentation problem followed by connected-component labelling, with a small residual need to split touching pairs. We therefore built a deliberately simple, fully reproducible pipeline around a semantic U-Net [7] and concentrated our effort on supervision, validation methodology and PQ-aware post-processing, measuring every design choice on out-of-fold data and keeping only what helped.

[1] https://github.com/cloudforge-oy/solar-filament-2026
[2] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Data preparation and targets

Images are read as single-channel 8-bit grayscale and scaled to $[0, 1]$; no other pre-processing (no limb-darkening correction, no disk masking) is applied. For every annotation set, the single-polygon segmentations are rasterised with `pycocotools` into a 2048×2048 instance map. The k annotation sets of one image are then reduced to three *consensus* targets that are shared by all annotators of that image:

(1) the soft filament mask $y_{\text{seg}} \in [0, 1]$, the fraction of annotators who marked the pixel. Training on this soft label rather than on one arbitrary annotator (or on the union) makes the network predict the *expected* annotation, which is the right target when the hidden test ground truth also consists of several annotator sets;
(2) a normalised distance-to-boundary map y_{dist}, computed per instance as the Euclidean distance transform divided by the instance maximum, averaged over annotators;
(3) a contact-line map y_{touch}: pixels within 2 px of two different instances of the same annotation set, averaged over annotators.

The two auxiliary targets were meant to separate touching filaments. The distance map feeds a marker-based watershed [3]; already on ground-truth maps, however, watershed over-segments elongated filaments, whose normalised distance has several maxima along the spine (oracle PQ 0.75 versus 0.79 for plain connected components on the same consensus masks). The contact-line map is non-zero only where two filaments meet, and cutting components along the predicted line is harmless, but with only ≈70 touching pairs in the training set the head learns little and the cut never changed validation PQ. Both heads were therefore kept as regularising auxiliary tasks, and the submitted instances come from plain connected components (Section 3).

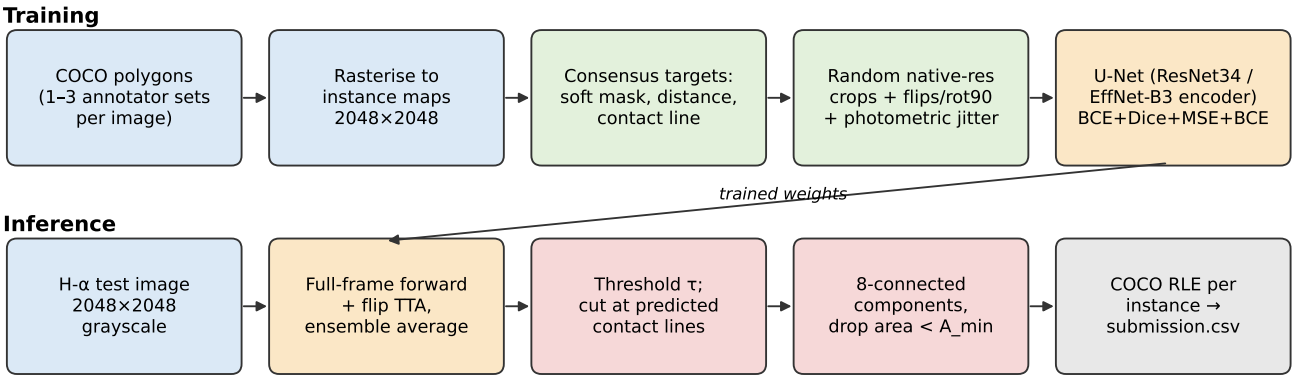


Figure 1: End-to-end pipeline. Training (top): COCO polygons of all annotator sets are rasterised into instance maps and reduced to three consensus targets per image; U-Nets are trained on random native-resolution crops. Inference (bottom): full-frame prediction with flip TTA, ensemble averaging, thresholding, connected components cut at predicted contact lines, minimum-area filtering, COCO RLE encoding.

2.2 Model

We use the U-Net implementation of `segmentation_models_pytorch` [5] with ImageNet-pretrained encoders, the first convolution adapted to one input channel, and three output channels (filament logit, distance logit, contact logit). Two encoders were compared: ResNet34 [4] (24.4 M parameters) and EfficientNet-B3 [8] (13.2 M parameters); the final submission uses EfficientNet-B3 only. The loss is

$$\mathcal{L} = \mathrm{BCE}(\hat{y}_{\mathrm{seg}}, y_{\mathrm{seg}}) + \mathrm{Dice}(\hat{y}_{\mathrm{seg}}, y_{\mathrm{seg}}) + 5\,\mathrm{MSE}(\hat{y}_{\mathrm{dist}}, y_{\mathrm{dist}}) + 0.5\,\mathrm{BCE}_{w=20}(\hat{y}_{\mathrm{touch}}, y_{\mathrm{touch}}),$$

where both BCE terms accept soft targets and the contact term uses a positive-class weight of 20 because contact pixels are extremely rare.

2.3 Sampling and training

Networks are trained on random crops at native resolution (512×512 for ResNet34, 640×640 for EfficientNet-B3), four crops per image per epoch; with probability 0.7 the crop is centred near a filament pixel of the consensus mask, otherwise it is placed uniformly at random (so that empty sky, limb and off-disk corners are seen). Augmentations are horizontal/vertical flips, 90° rotations, random gamma ($e^{\pm 0.3}$), contrast (±15%), brightness (±0.05) and Gaussian noise ($\sigma \le 0.02$). We use AdamW (weight decay 10^{-4}), a one-cycle learning-rate schedule peaking at $3 \cdot 10^{-4}$, mixed precision, batch size 8 (ResNet34, 40 epochs) or 6 (EfficientNet-B3, 50 epochs). One ResNet34 run takes about 45 minutes on a laptop RTX 5060 (8 GB).

Validation split. The 707 images are grouped by observation date and the dates are assigned to five folds, so that the same day observed by two GONG stations, or the same image annotated by three annotators, never straddles the training/validation boundary. Every configuration is trained five times, once per held-out fold, so that each of the 1154 annotation sets receives an out-of-fold prediction; all numbers in Section 3 are out-of-fold. The final ensemble averages the five EfficientNet-B3 fold models (each trained on 80% of the images), which uses all training data without ever tuning on data a model has seen.

2.4 Inference and post-processing

Each test image is processed as a whole 2048×2048 frame (0.4 s per forward pass in fp16). Predictions are averaged over the identity and three flips (TTA) and over the five fold models. Instances are then extracted as follows: (i) threshold the filament probability at τ; (ii) label 8-connected components; (iii) drop components smaller than $A_{\min}$ pixels; (iv) encode each component as a COCO RLE string. Because components are disjoint by construction, no two predictions of an image overlap, which the PQ scorer requires. τ and $A_{\min}$ are chosen by a grid search that maximises out-of-fold PQ (Section 3); the minimum area is the most important knob, because PQ charges 0.5 for every false positive and the probability maps contain many small spurious dark blobs. The optional contact-line cut (remove pixels with contact probability above τ_c, label, then return the removed pixels to the adjacent component with a flat watershed) is implemented and evaluated but disabled in the submission.

3 EVALUATION

3.1 Strategy

We re-implemented PQ exactly as defined by the organizers: predicted and ground-truth instances are matched by IoU > 0.5 (unique by construction), $\mathrm{PQ} = \sum_{TP} \mathrm{IoU} / (|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|)$, computed per image *and per annotation set* (an image annotated by three people yields three PQ values) and averaged. Scoring against each annotator separately rather than against a merged mask keeps the validation faithful to the hidden test protocol, and the spread between annotators of the same image gives a direct estimate of the ceiling imposed by label noise. We also report the pixel Dice of the

consensus mask (threshold 0.5). Uncertainty is given as the standard error of the mean over annotation sets and as 95% bootstrap intervals over images.

3.2 Results

Table 1 gives the out-of-fold PQ of every configuration, fold by fold and pooled over all 1154 annotation sets. Post-processing (τ, $A_{\min}$) is re-optimised per fold on a grid ($\tau \in \{0.4, \ldots, 0.7\}$, $A_{\min} \in \{250, 400, 550\}$); the chosen values move within $\tau \in [0.4, 0.7]$, $A_{\min} \in \{400, 550\}$ and the PQ surface is flat there (Figure 2), so the fixed setting $\tau = 0.6$, $A_{\min} = 400$ was used for the submission. The contact-line cut (any τ_c) and the distance-map watershed never exceeded plain connected components on any fold, and rejecting components by their peak or mean probability changed PQ by less than 0.002, so both were disabled. EfficientNet-B3 beats ResNet34 by 0.02–0.03 PQ on every fold, and averaging the two encoders is no better than EfficientNet-B3 alone, so the submitted ensemble consists of the five EfficientNet-B3 fold models only.

On the public Kaggle leaderboard (about half of the 180 test images) the submissions scored: single-channel ResNet34 baseline 0.33; EfficientNet-B3 fold-0 model 0.35; five-fold EfficientNet-B3 ensemble 0.35 (both $\tau = 0.6/A_{\min} = 400$ and $\tau = 0.5/A_{\min} = 500$). The leaderboard is reported with two decimals, so the small differences visible in cross-validation are below its resolution; the roughly 0.06 gap between cross-validation and leaderboard PQ is constant across models and is discussed below.

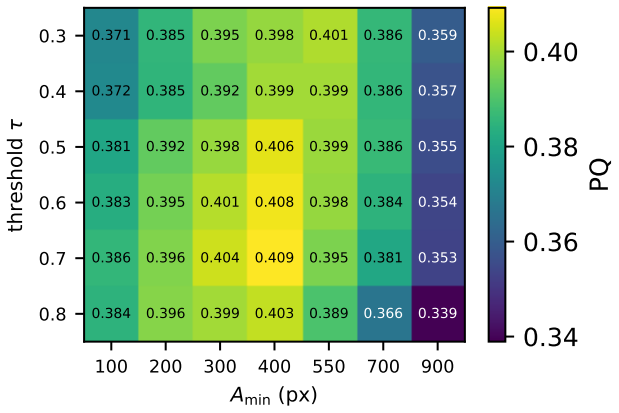


Figure 2: Out-of-fold PQ on fold 0 as a function of the probability threshold τ and the minimum instance area $A_{\min}$ (EfficientNet-B3, flip TTA). The optimum is broad.

3.3 Discussion

Strengths. The pipeline is small (five 13 M-parameter models), each model trains in about 50 minutes on an RTX 4080 or 2 hours on an 8 GB laptop GPU, inference runs at ≈1.5 s per full-resolution image per model including TTA, there is no learned instance machinery that could produce overlapping masks, and every design choice is backed by an out-of-fold number. Consensus supervision makes the probability map an estimate of annotator agreement, which is directly interpretable: faint filaments annotated by one of three annotators receive $p \approx 1/3$ and are correctly dropped at $\tau \geq 0.5$.

Weaknesses. False positives dominate the error budget (pooled FP ≈ 0.7×TP for EfficientNet-B3, Table 1): small dark features (sunspot penumbrae, dark fibrils, plage boundaries, seeing artefacts) are confused with short filaments, and the false positives are *confident*, which is why confidence filtering did not help. A sizeable part of them is annotator disagreement rather than model error: the same component is often a true positive against one annotator of an image and a false positive against another, and the consensus-mask oracle itself only reaches PQ 0.79 against individual annotators. The constant ≈ 0.06 gap between cross-validation and the public leaderboard most likely reflects a different aggregation of the hidden metric (e.g. dataset-level counts or a different treatment of multiply-annotated images) rather than overfitting, since it is identical for models of very different strength. Finally, connected components cannot split filaments that genuinely touch; with only ≈70 touching pairs in the training data the contact-line head did not learn a usable signal.

Possible improvements. A larger input context for faint, fragmented filaments (a lower-resolution auxiliary branch or a transformer encoder), explicit modelling of each annotator as a separate output head so that the mapping from consensus probability to PQ-optimal decisions can be learned instead of thresholded, a learned false-positive rejector on component-level features (area, darkness contrast, elongation, distance to the limb, proximity to sunspots), and spine-aware merging of fragments that belong to one filament. Longer schedules and larger encoders are the obvious compute-bound gains.

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Table 1: Out-of-fold Panoptic Quality (mean over annotation sets) per fold and pooled over all 1154 annotation sets (± standard error), consensus-mask pixel Dice, and matched instance counts summed over all folds. All models use flip TTA; post-processing is re-optimised per fold.

Model	fold 0	fold 1	fold 2	fold 3	fold 4	pooled PQ	Dice	TP / FP / FN
ResNet34 U-Net	0.386	0.354	0.398	0.382	0.373	0.378 ± 0.004	0.660	5001 / 4782 / 3198
EfficientNet-B3 U-Net	0.409	0.382	0.426	0.405	0.399	0.404 ± 0.004	0.681	5115 / 3702 / 3084
Both encoders (equal weights)	0.407	0.373	0.419	0.405	0.395	0.399 ± 0.005	0.677	5122 / 3998 / 3077

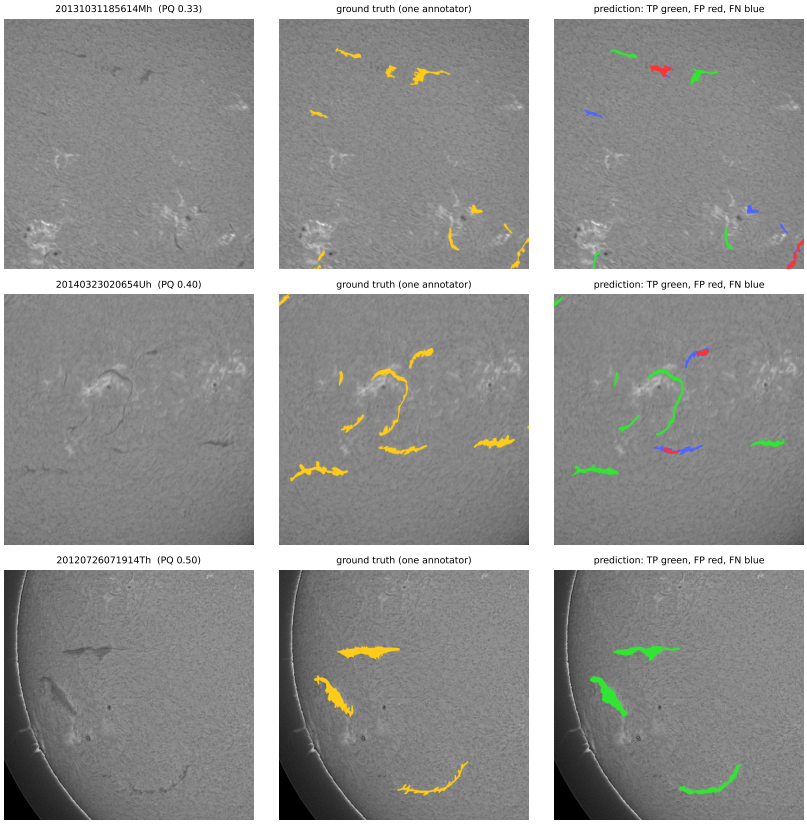


Figure 3: Validation images at the 25th, 50th and 75th percentile of per-image PQ (EfficientNet-B3 fold 0, out-of-fold). Left: H-α crop; middle: one annotator's ground truth; right: prediction with true positives in green, false positives in red and missed filaments in blue.

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